

A CLASS OF MULTI-PARAMETER POWER SERIES DISTRIBUTIONS

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SUMMARY

To supplement applications in cases where either the well-known power series distribution (PSD) or generalized PSD are usually applicable, a further extension of these distributions is considered in this paper. The probability generating function of this $m+1$ -parameter PSD with parameters $\theta_0, \dots, \theta_m$ is of the form:

$$M(t) = F(\sum_{i=0}^m \theta_i t^i) / F(\sum_{i=0}^m \theta_i)$$

where $F(\cdot)$ is the series function of a classical one-parameter power series distribution. This distribution can originate in various ways. A PSD which is generalized with a distribution having non-zero probabilities proportional to $\theta_0, \dots, \theta_m$ is one example.

Also the distribution of $\sum_{i=1}^m i R_i$ is of this form, assuming that $R_1, \dots, R_m$ have a sum-symmetric multivariate PSD with parameters $\theta_1, \dots, \theta_m$.

The moments and factorial moments of the distribution are obtained by means of a recursive relation. Attention is drawn to methods for estimating the parameters initially in order to obtain maximum likelihood estimators (m.l.e.) numerically with asymptotic expressions for the variances and covariances of these estimators. Special attention is paid to the binomial PSD for which m.l.e. are derived. Finally examples are given to illustrate the estimation and goodness of fit of empirical logseries distributions.

1. Introduction

It is well known that power series distributions containing the Poisson-, negative binomial-, binomial- and logarithmic series (logseries) distributions as special cases, play a useful part in the description of counts of organisms or individuals in the biological sciences. In some situations compound distributions with more than one parameter are necessary. Consider, for example, the classical problem where egg mass counts per square unit have a Poisson distribution and the larvae counts within an egg mass has a power series distribution (PSD). The compound or generalized Poisson in this case would be respectively a Neyman Type A-, negative binomial, Poisson-Binomial or Poisson-Pascal depending on whether this PSD is taken to be a Poisson, logseries, binomial or negative binomial distribution. Examples as well as properties of such distributions are found in Johnson and Kotz (1969), chapter 8.

Khartri and Patel (1961) considered three general classes of discrete distributions which contain most of the compound distributions mentioned in Johnson and Kotz (1969) and which are appropriate in situations such as the example of egg masses of larvae mentioned above.

Let us consider a class of distributions with probability generating function (p.g.f.) of the form

$$F(\sum_{i=0}^{\infty} \theta_i t^i) / F(\sum_{i=0}^{\infty} \theta_i),$$

where $F(\gamma) = \sum_{x=0}^{\infty} A_x \gamma^x < \infty$ and $A_x \geq 0$, some real function of x . Here $F(\cdot)$ is the series function of a PSD, while the θ_i 's are proportional to the probabilities of the compounding discrete distribution. We now specialize by considering a compounding distribution with finite, say $m+1$, points having positive probabilities. Therefore we define an m -parameter power series distribution by the p.g.f.

$$(1) \quad M(t) = F(\sum_{i=0}^m \theta_i t^i) / F(\sum_{i=0}^m \theta_i)$$

with θ_0 specifying the probability of the zero count in cases where a zero-truncated distribution is modified (see Johnson and Kotz, section 8.4) to include zero counts.

In this paper we proceed to study this general class of distributions with respect to its genesis in several situations, recurrence relations for moments and estimation of parameters. In the last two sections attention will be paid to a special case namely the binomial power series distribution, and some numerical examples.

2. The genesis of the distribution

The distributions with p.g.f. of the form (1) can be obtained by different methods in practice depending on the appropriate assumptions.

2.1 As a generalized PSD

In the example referred to in the previous section, the egg mass counts may be considered having a one-parameter PSD specified by $F(\cdot)$, while the larvae counts in egg masses can be taken as the generalizer-distribution with probabilities proportional to $\theta_0, \theta_1, \dots, \theta_m$ for counts $0, 1, \dots, m$.

2.2 As a univariate multinomial distribution compounded with a PSD

Consider a single trial with multiple events where either zero, one, two, ..., or m successes of an event can occur (e.g. a motor accident where either 0, 1, 2, ..., m deaths occur).

Let the probability of the i th success, $i = 0, 1, \dots, m$ be proportional to θ_i . If the number of independent repetitions of the trial is in itself a random variable having a one parameter PSD then the total number of successes has a m -parameter PSD. (This is the univariate analogue to property 4, Joshi (1975)).

2.3 From a multivariate PSD

Lemma 1:

Let $R_1, \dots, R_m$ have an m -variate PSD with parameters $\theta_1, \dots, \theta_m$ i.e.

$$P(R_1 = r_1, \dots, R_m = r_m) = A_{r_1, \dots, r_m} \theta_1^{r_1} \dots \theta_m^{r_m} / f(\theta_1, \dots, \theta_m)$$

where

$$f(\theta_1, \dots, \theta_m) = \sum_{\{r_1, r_2, \dots, r_m \geq 0\}} A_{r_1, \dots, r_m} \theta_1^{r_1} \dots \theta_m^{r_m}.$$

Suppose that for fixed $n = \sum_{j=1}^m r_j$,

$$(2) \quad A_{r_1, \dots, r_m} = \phi(n) / \prod_{j=1}^m r_j!,$$

with $\phi(n)$ a positive real function of n , then $\sum_{j=1}^m jR_j$ has a m -parameter PSD.

Proof:

The p.g.f. of $\sum_{j=1}^m jR_j$ can be written as

$$\begin{aligned} & \{r_1, \dots, r_m \geq 0\} A_{r_1, \dots, r_m} \prod_{j=1}^m \theta_j^{r_j} t^{\sum_{j=1}^m j r_j} / f(\theta_1, \dots, \theta_m) \\ &= \{r_1, \dots, r_m \geq 0\} \phi(n) / n! \{n! \prod_{j=1}^m \theta_j^{r_j} / r_j!\} t^{\sum_{j=1}^m j r_j} / f(\theta_1, \dots, \theta_m) \\ &= \sum_{n=0}^{\infty} \phi(n) / n! (\sum_{j=1}^m \theta_j t^j)^n / f(\theta_1, \dots, \theta_m). \end{aligned}$$

Since $f(\theta_1, \dots, \theta_m) = \sum_{n=0}^{\infty} \phi(n) / n! (\sum_{j=1}^m \theta_j)^n$, this p.g.f. is of form (1), with $\theta_0 = 0$.

Remarks:

(a) Among the multivariate power series distributions which have the property (2) are

the multinomial with $\phi(n) = n!$; the multivariate Poisson with $\phi(n) = 1$; the negative multinomial with $\phi(n) = (k+n)! / k!$ (k a fixed parameter), and the multivariate logseries with $\phi(n) = (n-1)!$, $n \geq 1$. (See also Van Heerden (1961)).

- (b) The multivariate PSD distributions having property (2), form the class of sum-symmetric PSD-distributions since $f(\theta_1, \dots, \theta_n) = g(\sum_{i=1}^n \theta_i)$ for some $g(\cdot)$ (cf. Joshi (1975)).

3. Some properties of the distributions

In estimation of the parameters it is more convenient only to estimate m parameters. This can be done by reparameterizing (1) such that it is of the form:

$$(3) \quad M(t) = \omega + (1-\omega)F(\sum_{j=1}^m \theta_j^* t^j) / F(\sum_{j=1}^m \theta_j^*)$$

with $0 < \omega < 1$ some function of $\theta_0, \dots, \theta_m$, giving the probability of zero counts in cases where $F(0) = 0$ (when modifying the distribution).

When $F(0) > 0$, we can take $\omega = 0$ without loss in generality. In this form estimation of $\theta_0, \dots, \theta_m$ is reduced to the estimation of $\theta_1^*, \dots, \theta_m^*$ from

$$(4) \quad F(\sum_{j=1}^m \theta_j^* t^j) / F(\sum_{j=1}^m \theta_j^*)$$

The probability of the r th count can be obtained from $\{M^{(r)}(t)\}_{t=1}, r = 1, 2, \dots$ (Here $M^{(r)}(t)$ is the r th derivative of $M(t)$ with respect to t).

A recurrence relation in moments can be obtained by introducing "exponential moments". Define the factorial moment generating function (f.m.g.f.) as

$$(5) \quad G(\alpha) = M(1 + \alpha).$$

Also define $H(\alpha) = \exp \{G(\alpha) - 1\}$.

Now, by expanding $H(\alpha)$ in the form $(\mu_{\{1\}}\alpha + \mu_{\{2\}}\alpha^2/2! + \dots)$, the coefficient of $\alpha^i/i!$ gives the i th exponential moment.

Write $g(\alpha) \equiv g(\alpha|\theta_0, \dots, \theta_m) = \sum_{x=0}^{\infty} A_x \sum_{r=1}^x \binom{x}{r} (\sum_{j=0}^m \theta_j^*)^r (\sum_{j=0}^m \theta_j^* \sum_{i=1}^j \binom{j}{i} \alpha^i)^{x-r}$ and $c = 1/F(\sum_{j=0}^m \theta_j^*)$, then it is apparent that $H(\alpha) = \exp \{cg(\alpha)\}$.

For the function $H(\alpha)$ of this form the following can be written down generally:

$$(6) \quad H^{(r)}(\alpha) = c \{ \sum_{i=1}^r a_{r-1,j} g^{(i)}(\alpha) H^{(r-i)}(\alpha) + \sum_{i=2}^{r-1} \binom{r-2}{i-2} g^{(i)}(\alpha) H^{(r-i)}(\alpha) \}$$

where $a_{11} = a_{22} = 1$; $a_{i1} = 1$; $a_{ii} = 2$ for $i = 3, 4, \dots$; $a_{ij} = a_{i-1, j-1} + a_{i-1, j}$ for any other $j < i$.

By noting that $\mu_{\{r\}} = H^{(r)}(0)$, the recurrence relation in terms of exponential moments is:

$$(7) \quad \mu_{\{r\}} = c[\sum_{i=1}^r a_{r-1, i} g^{(i)}(0) \mu_{\{r-i\}} + \sum_{i=2}^{r-1} a_{i-2}^{r-2} g^{(i)}(0) \mu_{\{r-i\}}]$$

with $\mu_{\{0\}} = 1$.

From the definition of $H(\alpha)$, relations between exponential and factorial moments can easily be written down:

$$\begin{aligned} \mu_{\{1\}} &= \mu_{(1)} = \mu'_1 \\ (8) \quad \mu_{\{2\}} &= \mu_{(2)} + \mu_{(1)}^2 \\ \mu_{\{3\}} &= \mu_{(3)} + 3\mu_{(1)}\mu_{(2)} + \mu_{(1)}^3 \\ \mu_{\{4\}} &= \mu_{(4)} + 4\mu_{(1)}\mu_{(3)} + 3\mu_{(2)}^2 + 6\mu_{(1)}^2\mu_{(2)} + \mu_{(1)}^4 \text{ etc.} \end{aligned}$$

We now write down the first two factorial moments of the m-parameter PSD:

$$\begin{aligned} \mu_{(1)} &= g'(0|\theta_0, \dots, \theta_m)/F(\sum_{i=0}^m \theta_i) \\ &= \sum_{x=0}^{\infty} x A_x(\sum_{i=0}^m \theta_i)^{x-1} / F(\sum_{i=0}^m \theta_i) \\ &= (\sum_{i=1}^m i\theta_i / \sum_{i=0}^m \theta_i) \mu_{(1)}^{\star}(\sum_{i=0}^m \theta_i), \end{aligned}$$

where $\mu_{(k)}^{\star}(\lambda)$ is the k'th factorial moment of the one parameter PSD with parameter λ and series function $F(\lambda)$, for $k = 1, 2, \dots$.

From (7) and (8) follows

$$\mu_{\{2\}} = [g^{(1)}(0) \mu_{\{1\}} + g^{(2)}(0)] / F(\sum_{i=0}^m \theta_i),$$

and therefore

$$\mu_{(2)} = (\sum_{i=2}^m i(i-1)\theta_i / \sum_{i=0}^m \theta_i) \mu_{(2)}^{\star}(\sum_{i=0}^m \theta_i) + (\sum_{i=1}^m i\theta_i / \sum_{i=0}^m \theta_i)^2 \mu_{(1)}^{\star}(\sum_{i=0}^m \theta_i).$$

4. Estimation of the parameters

In general it is only possible to obtain maximum likelihood estimators for the parameters of the m -parameter PSD numerically by evaluating the log-likelihood function on a grid of combinations of parameter-values. In order to start the searching process, crude estimates for the parameter are necessary. Among various possible types of estimates, we shall concentrate mainly on methods which are based on frequencies and moments:

4.1 Frequency estimates:

(a) **Case I:** $F(0) = 0$; distribution truncated at zero:

From (3), writing

$$(9) \quad \delta = \sum_{i=1}^m \theta_i,$$

we obtain in terms of the probabilities p_i of count $i = 1, 2, \dots$,

$$(10) \quad p_1 = F^{(1)}(0)\theta_1/F(\delta),$$

$$(11) \quad f_2 = p_2/p_1 = F^{(2)}(0) \cdot \frac{1}{2}\theta_1/F^{(1)}(0) + \theta_2/\theta_1 \text{ etc.}$$

Choosing some initial value $\delta^{(0)}$ for δ , ($0 < \delta^{(0)} < 1$), $\theta_1^{(0)}$ is obtained from (10) by using the observed frequency of count 1. Then $\theta_2^{(0)}$ is the solution of (11), etc. The values of $\theta_i^{(0)}$ $i = 1, \dots, m$ when substituted in (9) give $\delta^{(1)}$ which can be used to obtain $\theta_1^{(1)}, \theta_2^{(1)}, \dots, \theta_m^{(1)}$.

This iteration process can be terminated as soon as convergence in the sequence $\{\delta^{(r)}\}$ is apparent. Divergence calls for a better initial guess of δ .

(b) **Case II:** $F(0) = 0$; distribution modified, by taking $\theta_0 > 0$:

Here we have

$$p_0 = F(\theta_0)/F(\sum_{i=0}^m \theta_i)$$

$$(12) \quad f_1 = p_1/p_0 = F^{(1)}(\theta_0)\theta_1/F(\theta_0)$$

$$(13) \quad f_2 = F^{(2)}(\theta_0) \cdot \frac{1}{2}\theta_1/F^{(1)}(\theta_0) + \theta_2/\theta_1.$$

With an initial value $\theta_0^{(0)}, \theta_1^{(0)}$ and $\theta_2^{(0)}$ are, as before, the solutions of (12) and (13) etc.

By using the Newton-Raphson method and taking

$$\theta_0^{(1)} = \theta_0^{(0)} - R(\theta_0^{(0)})/R'(\theta_0^{(0)})$$

where $R(\theta_0^{(0)}) = p_0 F(\sum_{i=0}^m \theta_i^{(0)}) - F(\theta_0^{(0)})$, and from it $\theta_1^{(1)}$ and $\theta_2^{(1)}$ etc., the iteration process leads quickly to a solution provided that the initial estimate of θ_0 is not too crude.

(c) **Case III:** $F(0) > 0$.

In this case

$$p_0 = F(0)/F(\delta), \text{ so that}$$

$$f_1 = F^{(1)}(0)\theta_1/F(0),$$

from which θ_1 can be estimated by taking f_1 as the ratio of the observed frequencies of counts 1 and 0. From (11) θ_2 is obtained, etc.

4.2 Moment estimates

Writing $\delta = \sum_{i=1}^m \theta_i$, $c_j = \sum_{i=j}^m i^{(j)} \theta_i$, $j = 1, \dots, m$; where $i^{(j)} = i(i-1) \dots (i-j+1)$,

then by using the fact that $\mu_{(i)} = \frac{d^i}{dt^i} M(t)|_{t=1}$ and using the form in (3):

$$(14) \quad \mu_{(1)} = [F^{(1)}(\delta) \cdot c_1 / F(\delta)](1-p_0)$$

$$(15) \quad \mu_{(2)} = [(F^{(2)}(\delta)c_1^2 + F^{(1)}(\delta)c_2)/F(\delta)](1-p_0) \text{ etc.,}$$

where $p_0 > 0$ for distributions where $F(0) = 0$ and are modified to include zero counts.

An initial value for δ , $\delta^{(0)}$ say, substituted into (14) where $\mu_{(1)}$ is taken to be the observed factorial moment (mean), and p_0 to be the relative frequency of zero, gives a solution, $c_1^{(0)}$ say, for c_1 . The values of $\delta^{(0)}$ and $c_1^{(0)}$ in (15) give $c_2^{(0)}$ in terms of the observed second factorial moment. This process is repeated until $c_m = m!$. θ_m is expressed in terms of $\mu_{(1)}, \dots, \mu_{(m)}$ and $\delta^{(0)}$. Now $\theta_{m-1}^{(0)}$ can be obtained from

$$c_{m-1}^{(0)} = (m-1)! \theta_{m-1}^{(0)} + m! \theta_m^{(0)}$$

and by substitution the values of $\theta_{m-2}^{(0)}, \dots, \theta_1^{(0)}$ are obtained. by taking $\delta^{(1)} = \sum_{i=1}^m \theta_i^{(0)}$, a second step in the iteration process commences. The process is terminated when it is

apparent that the sequence $\{\delta^{(r)}\}_{r=0, 1, \dots}$ converges.

A combination of the frequency and moment methods of estimation is another possibility.

4.3 Moment-least squares estimates

Using expressions (14) and (15) it is also possible to express the non-central moments μ'_1 and μ'_2 in terms of δ , c_1 and c_2 (which are functions of θ_1 and θ_2). In the same way $\mu'_3, \dots, \mu'_m$ can be expressed in terms of $\delta, c_1, \dots, c_m$. We now proceed in the same way as Steyn (1976) by letting $Y_i = m'_i - \mu'_i, i = 1, \dots, m$, where m'_i is the observed i 'th non-central moment and using the fact that the moments are correlated.

5. The Binomial-power series distribution

This distribution most naturally arises when the parameter n of a binomial distribution is a random variable for which n/m follows a PSD. The p.g.f. is:

$$M(t) = F(\lambda(1-p+pt)^m)/F(\lambda),$$

This distribution is a special case of the m -parameter PSD with

$$\theta_0 = \lambda(1-p)^m$$

$$\theta_i = \lambda \binom{m}{i} p^i (1-p)^{m-i}, i=1, \dots, m$$

in (1) and

$$\theta_i^* = \theta_i / (1 - \lambda(1-p)^m), i=1, \dots, m;$$

$$\omega = F(\theta_0)/F(\lambda)$$

in (3).

The probability function is given by

$$P_x = \sum_{j \geq x/m}^{\infty} A_j \lambda^j \binom{mj}{x} p^x q^{mj-x} / F(\lambda), q=1-p.$$

The maximum likelihood estimates of this distribution can be derived as follows:

Following the same lines as Spratt (1958), with

$$S_1(x) = F(\lambda)P_x$$

it is easy to show that

$$(16) \quad \frac{\partial}{\partial \lambda} S_1(x) = [(x+1)qS_1(x+1)/p + xS_1(x)]/m\lambda$$

and

$$(17) \quad \frac{\partial}{\partial p} S_1(x) = [xS_1(x) - (x+1)S_1(x+1)]/p$$

Differentiating the log-likelihood function with respect to λ and p and using (16) and (17), we have

$$(18) \quad [(q/p)\sum_{x=0}^k n_x(x+1)S_1(x+1)/S_1(x) + n\bar{x}]/m\lambda = nF'(\lambda)/F(\lambda)$$

and

$$(19) \quad n\bar{x} = \sum_{x=0}^k n_x(x+1)S_1(x+1)/S_1(x)$$

where $n_0, \dots, n_k$ are the observed frequencies, $\sum_{i=0}^k n_i = n$ and $\bar{x} = \sum_{i=1}^n i n_i / n$. Substitution of (19) into (18) results in

$$(20) \quad \bar{x} = F'(\lambda) \cdot m\lambda p / F(\lambda)$$

The likelihood estimates can be obtained iteratively from (19) and (20) following the method of Shenton (1949):

Writing $S_1(x, \lambda)$ for $S_1(x)$ when (20) is used to express it as a function of λ only, and writing

$$G(\lambda) = \sum_{x=0}^k n_x(x+1)S_1(x+1, \lambda)/S_1(x, \lambda) - n\bar{x}$$

one obtains

$$\begin{aligned} G'(\lambda) = & \left[\sum_{x=0}^k n_x(x+1)S_1(x+1, \lambda)/S_1(x, \lambda) \right. \\ & + (q/p)\sum_{x=0}^k n_x(x+1)(x+2)S_1(x+2, \lambda)/S_1(x, \lambda) \\ & \left. - (q/p)\sum_{x=0}^k n_x(x+1)^2 S_1^2(x+1, \lambda)/S_1^2(x, \lambda) \right] / m\lambda \end{aligned}$$

By the Newton-Raphson method, iteratively,

$$\lambda^{(r)} = \lambda^{(r-1)} - G(\lambda^{(r-1)})/G'(\lambda^{(r-1)}), \quad r = 1, 2, \dots$$

by using an initial value $\lambda^{(0)}$, obtained by one of the methods in section 4.

The asymptotic variance-covariance matrix of the maximum likelihood estimates $\hat{\lambda}$ and $\hat{p}$ is the inverse of the matrix

$$\begin{bmatrix} I_{\lambda\lambda} & I_{\lambda p} \\ I_{\lambda p} & I_{pp} \end{bmatrix}$$

where

$$\begin{aligned} I_{\lambda\lambda}/n &= E\left(\left(\frac{\partial}{\partial\lambda} P_x/P_x\right)^2\right) \\ &= (F'(\lambda))^2/F^2(\lambda) + [\mu_{(1)}(1/m\lambda \\ &\quad - 2F'(\lambda)/pF(\lambda)) + (1+2q/p)\mu_{(2)}]/m\lambda \\ &\quad + (q^2/p^2 m\lambda)\phi]/m\lambda \end{aligned}$$

$$\begin{aligned} \text{and } \phi &= E\{(x+1)^2 S_1^2(x+1)/S_1^2(x)\} \\ &\doteq \sum_{x=0}^k (x+1)^2 p_{x+1}^2 / p_x, \end{aligned}$$

$$\text{if } \sum_{x=k+1}^{\infty} (x+1)^2 p_{x+1}^2 / p_x \text{ is negligible.}$$

Also

$$\begin{aligned} I_{pp}/n &= (\mu_{(1)} - \mu_{(2)} + \phi)/p^2, \\ \text{and } I_{\lambda p}/n &= [(q/p)(\mu_{(2)} - \phi) + \mu_{(1)}]/pm\lambda \end{aligned}$$

Special distributions:

By choosing $F(\lambda)$ appropriately, special cases of the binomial PSD are obtained:

(a) Binomial-Poisson distribution

This distribution is described in detail in Johnson and Kotz pp. 190-194 under the name Poisson-Binomial.

(b) Binomial-logseries distribution:

Very little literature is available on this distribution (see Johnson and Kotz, p. 195

also Plunkett and Jain (1975)). It will be considered in more detail in the examples of section 6.

(c) Binomial-binomial distribution:

Except in the case where one (n) of the three parameters is known, this distribution has few applications (see Johnson and Kotz, p. 194).

(d) Binomial-negative-binomial distribution:

For the same reason as in (c) this distribution has applications mostly when the parameter k of the negative binomial is known. (See Johnson and Kotz, p. 194-195).

Charalambides (1977) gives recursion-formulas for the probabilities and factorial moments of (a) – (c) above. (cf. also Khatri and Patel (1961)).

6. Examples

Example 1

Kemp and Kemp (1956) used Student's original data on haemocytometer counts of yeast cells together with expected frequencies by the Poisson, Negative binomial, Neyman type A and Polya-Aeppli to compare with the fit by a Hermite distribution (a Binomial-Poisson distribution, with the parameters chosen appropriately).

Also appropriate to this example would be other binomial PSD's such as the binomial logseries distribution (BLSD). To illustrate the methods of estimation we fitted this distribution using the following methods (see Table 1):

(a) Frequency estimates:

We use the method of section 4.1(b) since $F(0) = 0$, but it is modified to include zero counts. Here the estimates are found to be $\check{\theta}_0 = 0,5547$, $\check{\theta}_1 = 0,2164$ and $\check{\theta}_2 = 0,0100$. Since $\theta_0 = \lambda(1-p)^2$, $\theta_1 = 2\lambda p(1-p)$ and $\theta_2 = \lambda p^2$, we have that $\lambda = \theta_0 + \theta_1 + \theta_2$, while $p = 2\theta_2/(\theta_1 + 2\theta_2)$, so that

$$\check{\lambda} = 0,7811 \text{ and } \check{p} = 0,0846$$

(b) Moment estimates:

Using section 4.2, with the first two observed factorial moments as $m_{(1)} = 0,682$ and $m_{(2)} = 0,595$ and taking $p_0 = 0,5325$, we have $\hat{\theta}_1^* = 0,2345$ and $\hat{\theta}_2^* = 0,0644$ as estimates of θ_1^* and θ_2^* (in (3)). The χ^2 -value is significant at a 5 percent level.

(c) Maximum likelihood estimates:

Following the method described in section 5, we have that $\bar{x} = 0,682$, $\hat{\lambda} = 0,6851$,

$\hat{p} = 0,1811$. These parameter estimates fit the binomial-logseries distribution rather well on the data (see Table 1) with a $\chi^2 = 4,88$ ($0,1 < P < 0,2$, with 3 degrees of freedom) compared to $\chi^2 = 3,88$ for the Hermite distribution. For the lower counts this distribution gives a better fit than the Hermite or any of the other distributions considered by Kemp and Kemp.

TABLE 1

No. of yeast cell per square	Observed frequency	Hermite	Expected freq.-Binomial-logseries		
			Frequency estimates	Moment estimates	Maximum likelihood estimates
0	213	213,0	213,0	213,0	212,9
1	128	123,0	128,0	123,5	130,1
2	37	46,7	37,0	48,4	38,8
3	18	13,3	13,0	10,2	11,5
4	3	3,1	5,1	3,4	4,1
5	1	0,6	2,1	1,0	1,5
6+	0	0,3	1,8	0,5	1,1

$$\begin{array}{lll} \hat{\lambda}=0,7811 & \hat{\theta}_1^*=0,2345 & \hat{\lambda}=0,6851 \\ \hat{p}=0,0846 & \hat{\theta}_2^*=0,0644 & \hat{p}=0,1811 \end{array}$$

The asymptotic variance-covariance matrix can be evaluated by taking $\phi \doteq 0,5059$ (assuming $k = 6$): $\text{Var}(\hat{\lambda}) \doteq 0,005621$, $\text{Var}(\hat{p}) \doteq 0,0004671$, $\text{Cov}(\hat{\lambda}, \hat{p}) \doteq -0,001360$.

Example 2

The size of free-forming groups of people were observed at the Garlicks Restaurant, Pretoria (cf. Steyn and Gerntholtz (1968)), resulting in the frequency table given in Table 2 (first two columns). Theoretically under certain assumptions, sizes of free-forming groups ought to have a truncated Poisson distribution (cf. Steyn and Gerntholtz), but for this situation the fit was poor, as is apparent from Table 2. Also, a logseries distribution (motivated by a possible non-constant Poisson parameter) turns out to be no answer.

Another approach is to assume that the group size has a truncated (at zero) binomial distribution with parameters p and n . In this case $n = 7$ seems feasible. Estimating p from the moment equation

$$\mu'_{(1)} = np/(1-p)^n,$$

by substituting the mean group size in the sample for μ'_1 and solving iteratively for p , we obtain an estimate $p^\star = 0,195$. The theoretical frequencies when using this distribution, are displayed in Table 2 and although the fit shows a significant difference ($\chi^2 = 9,577$ with two degrees of freedom), it points towards a better fit than the truncated Poisson distribution (with $\chi^2 = 14,47$ for three degree of freedom).

TABLE 2

Group size (x)	Observed Frequency (f)	Expected frequencies						
		Poisson	Binomial	Logseries	2-parameter logseries			
					$\tilde{\theta}_1, \tilde{\theta}_2$	$\check{\theta}_1, \check{\theta}_2$	$\tilde{\theta}'_1, \tilde{\theta}'_2$	$\check{\theta}'_1, \check{\theta}'_2$
1	136	151,5	142	187	132	136	139	140
2	123	93,5	103	60	134	123	126	122
3	27	38,5	42	26	18	22	19	21
4	9	12	10	12	11	12	10	11
5	2	3	2	6	3	4	3	3
6	1	0,5	0	3	1	2	1	2
7+	1	0	0	5	0	0	1	0

The assumption of a fixed n may be the reason for the poor fit in this example. We therefore proceed by investigating a compound binomial distribution with n/m having some truncated discrete distribution. Choosing as illustration the logseries distribution as compounder, it follows from section 2.2 that the truncated binomial-logseries distribution with $m = 2$ can be considered as a two-parameter logseries distribution. We now proceed to fit this distribution by using the various methods of estimation of the parameters.

(a) Frequency estimates:

Using the method described in section 4.1(a) we obtain the estimates $\check{\theta} = 0,1815$ and $\tilde{\theta} = 0,1476$. The only problem is that the iteration process as proposed converges only for a rather good choice for $\delta^{(0)}$. The fit when using these estimates is excellent as can be seen from Table 2 ($\chi^2 = 2,53$).

(b) Moment estimates:

To use this method as described in section 4.2, we need to calculate the first two sample factorial moments: $m_{(1)} = 1,746$ and $m_{(2)} = 2,1002$. An initial estimate for δ ,

can be obtained by using the value $m_{(1)}$ in the table of Patil (1962): $\delta^{(0)} = 0,64$. Using this as starting value, the iteration process as described in section 4.2 results in $\tilde{\delta} = 0,27$, $\tilde{\theta}'_1 = 0,1387$, $\tilde{\theta}_2 = 0,1312$. From Table 2 it is obvious that the two-parameter distribution using moment estimates gives a far better fit than the one-parameter case ($\chi^2 = 5,79$).

(c) Combined moment and frequency estimates:

Using (11) with $m = 2$ and the ratio of (14) and (15) simultaneously, it is possible to solve for θ_1 and θ_2 directly in terms of f_2 and $m_{(2)}/m_{(1)}$. We obtain $\tilde{\theta}'_1 = 0,1536$ and $\tilde{\theta}'_2 = 0,1312$. The fit with these estimates for the parameters is again good ($\chi^2 = 3,77$).

(d) Maximum likelihood estimates (m.l.e.):

Using the estimates $\tilde{\theta}'_1$ and $\tilde{\theta}'_2$ to give an indication of the region in which to search for the m.l.e., we chose the values $\theta_1 = 0,11(0,01)0,20$ and the same for θ_2 , and evaluated $\log L_n$ for every combination (θ_1, θ_2) . The maximum value was found at $(\theta_1, \theta_2) = (0,18; 0,14)$. With these values as basis, evaluations for $\log L_n$ were done for $\theta_1 = 0,171(0,001)0,190$ and $\theta_2 = 0,131(0,001)0,150$, for which the maximum value was at $(\theta_1, \theta_2) = (0,175; 0,137)$.

Taking these values as approximations to $\hat{\theta}_1$ and $\hat{\theta}_2$, it follows from Table 2 that these estimates provide a good fit, and indeed as could be expected asymptotically, have the smallest χ^2 -value, namely 2,4 [see Cramér (1946), the χ^2 -minimum method of estimation]. We used the $\hat{\theta}_1$ and $\hat{\theta}_2$ -values to estimate the asymptotic variances and covariance: $\text{Var}(\hat{\theta}_1) = 0,000844$, $\text{Var}(\hat{\theta}_2) = 0,000293$, $\text{Cov}(\hat{\theta}_1, \hat{\theta}_2) = 0,000260$. From the asymptotic normality of $\hat{\theta}_1$ and $\hat{\theta}_2$ e.g. it is quite clear that $\hat{\theta}_2$ differs significantly from zero.

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